Take-Home Exam, Advanced Microeconomics

Problem 1: Overworked Americans – Q1, Q2, Q4, Q5, Q7

Alessandro Caggia
Bocconi

May 2026

Plan and storyline. Q1 – there is an inconsistency in the data. The US–EU hours-per-worker gap is real (panels (a)/(b)) but its cross-country relation to the tax wedge is loose: the 2023 OLS slope is signed correctly (-9.9 hrs/pp) but $R^2 = 0.16$ and the residual SD is ≈ 158 hours – one third of the entire DE–US gap (panel (c)). Taxes account for some of the cross-country variation, not most of it. **Q2 – the elasticity needed to close the gap is larger than the literature delivers.** The Prescott log–log model gives $h^*(\tau) = (1-\tau)/(\alpha + 1 - \tau)$. Calibrating α to US hours and matching the cross-country gap requires a Marshallian elasticity of ~ 1, while Chetty–Guren–Manoli–Weber (2011) report a Hicksian intensive of ~ 0.30. The gap between what the model needs and what the micro evidence delivers is what we ask §4–§6 to absorb – something is constraining hours *beyond* the price of leisure. **Q4 – hours regulation:** a binding cap $\bar{h}$ is welfare-improving iff $\bar{h} \in (2h^{\text{soc}} - h^{\text{firm}}, h^{\text{firm}})$ in the firm-power baseline; below h^* it overshoots. **Q5 – single union:** bargaining weight η delivers $h(\eta)$ from h^{firm} ($\eta = 0$) to h^* ($\eta = 1$); joint surplus is single-peaked at an interior η^{soc} – a union can be welfare-improving or welfare-reducing depending on its strength. **Q6 – culture:** a higher leisure preference α^{EU} shifts h^* down at every τ; welfare-neutral by construction. **Q7 – welfare comparison and counterfactuals.** Shutting down EU's cultural shift counterfactually adds $\Delta W = +0.088$; shutting down its union channel *removes* $\Delta W = -0.060$. Net EU–US welfare difference at baseline is -0.028, real but quantitatively second-order to non-labour channels (Jones–Klenow 2016). Reproducible: `research/build_figure_panel.py`, `research/sim_mechanisms.py`.

1. Q1 – The hours gap and its noisy relation to tax wedges

Question 1. *Document the difference between hours worked in the US vs Europe in recent years.*

1.1 Definitions (used throughout the problem)

Three working-age (15–64) quantities, common notation:

- HE = *hours per employed worker* (annual hours actually worked; OECD `DSD_HW`). The intensive object the labour-supply model speaks to.
- $e \equiv E/N$ = *employment rate* (fraction of the working-age population that is employed; OECD `DSD_LFS`).
- $H/N \equiv HE \cdot e$ = *hours per adult* (total labour input per working-age person).

The body of the answer focuses on HE – the object that responds to the price of leisure in the static labour-supply model of Q2 and the firm–worker interaction model of Q4–Q6. The participation/extensive margin e is documented in this section but not analytically pursued in the body.[1]

[1] **Why we omit the extensive margin from the body.** Cross-country differences in e are substantial (US 71% vs IT 59% in 2019, BBF 2019 Tab. 2) and matter for total labour input H/N. We do not model them because (i) the participation decision is shaped by joint-vs-individual income tax filing, public childcare, employment-protection legislation and the structure of the welfare system – a separate body of theory to the labour-supply model that Q2 asks for; (ii) layering a participation margin on top of the firm-profit-max framework of Q4–Q6 would dilute the analytic focus of those questions without changing their qualitative welfare conditions; (iii) the question prompt asks specifically about *hours worked*, which the OECD reports unambiguously as HE. The relevant numerical evidence is documented in `research/repl_bbf_decomp.py` for the interested reader.

1.2 The fact

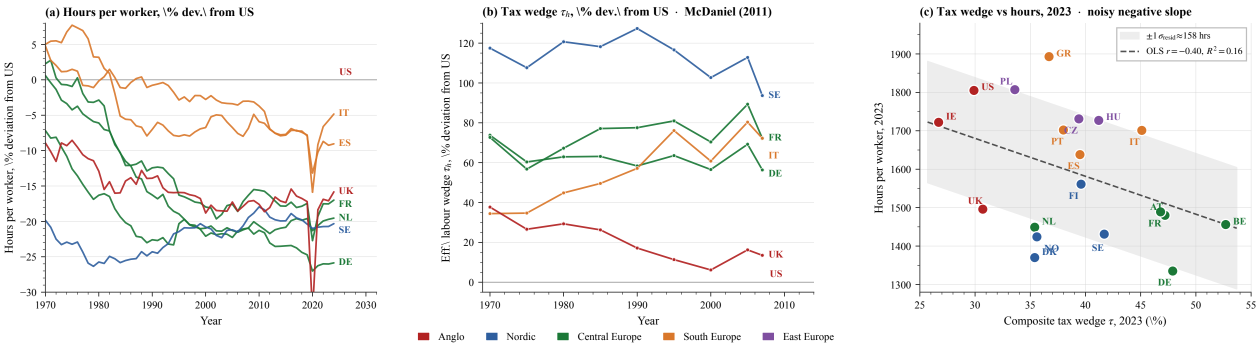

Figure 1: **Three-panel evidence.** (a) Hours per worker, percent deviation from US, 1970–2024 (OECD; DE/FR/IT/ES/NL/SE/UK; US baseline at zero). (b) Effective labour wedge τ_h, percent deviation from US, 1970–2007 (McDaniel 2011 national-accounts measure). (c) Composite tax wedge τ vs hours per worker, 2023 cross-section ($n = 19$ OECD countries); OLS line and $\pm 1\sigma_{\text{resid}}$ band. *Source / replication:* `research/build_figure_panel.py`.

Three readings of Fig. 1.

(i) Hours fell in every focus country relative to the US since 1970 (panel a). Northern Europe (DE, NL) is now $\approx -25\%$ to -30% below the US line; Southern Europe (IT, ES) at -10% to -15%. The gap widened steadily; it is not a recent phenomenon.

(ii) The labour wedge rose in every focus country relative to the US (panel b). Sweden is $+120\%$ above the US wedge in 2007; DE/FR/IT are $+60$ to $+70\%$. The two series move in opposite directions, qualitatively consistent with a tax-driven story.

(iii) The 2023 cross-section is signed correctly but very noisy (panel c). OLS slope $\hat{\beta} = -9.9$ hrs/pp, $r = -0.40$, $R^2 = 0.16$. The residual SD is $\approx$ 158 hours. To put it in scale: the entire DE–US hours gap is 470 hours, so the residual SD is roughly one third of the gap I am trying to explain. Italy ($\tau = 45.1\%$) and Greece ($\tau = 36.7\%$) work *more* hours per worker than France ($\tau = 46.8\%$); the four highest-τ countries (BE, DE, AT, FR) span $\approx$ 200 hours within a 6-pp wedge band.

1.3 The inconsistency the rest of the problem is asked to explain

The within-country time path (panels a/b) is consistent with a tax-driven story; the cross-country cross-section (panel c) is not. Either the elasticity that fits the time series is much larger than the cross-section delivers (the Prescott–CGM puzzle, Q2), or the cross-section is corrupted by country-specific institutional and cultural frictions that the tax wedge does not capture (Q4–Q6). Q2 formalises the first horn; §4–§6 build the second.

2. Q2 – Prescott model and the elasticity that would be needed

Question 2. *Summarise the main differences in income tax between US and Europe. Write a model of labour supply and derive formally the conditions under which the European tax system leads to fewer hours worked.* Use the words 'hence' and 'novel' twice.

2.1 Modelling choice and its trade-offs

Trade-off captured. I want a closed-form expression for $h^*(\tau)$ and a clean elasticity comparative static, so I can compare the model's prediction to the cross-section in panel (c) of Fig. 1 and to the micro estimates in the elasticity literature. Log–log preferences with a balanced lump-sum rebate are the standard choice (Prescott 2004; Cooley–Prescott 1995): they give a one-parameter (α) closed-form labour-supply function and a Marshallian elasticity that is monotone in α. **What I am ignoring (and why).** (i) The intertemporal margin: a static model has no Frisch elasticity, so I cannot speak to the business-cycle volatility of hours. This is fine because the question is about long-run cross-country differences, not high-frequency adjustments. (ii) A non-balanced budget: I assume the government rebates the full tax revenue lump-sum; relaxing this changes the income effect but not the substitution effect, leaving the qualitative comparative static intact. (iii) Capital and savings: log preferences in a static model imply consumption equals labour income. With positive savings the relevant wedge would be the wedge on consumption *plus* the wedge on capital income; for prime-age workers near the median the capital channel is small (CGM 2011 §3) so the omission is not first-order. (iv) Heterogeneity: the representative household assumption hides distributional effects, which I address in Q4 (heterogeneous workers under a uniform cap).

2.2 The relevant tax object: the composite wedge

Workers supply labour only to consume; *hence* a consumption tax is isomorphic to a labour tax in any static model with no untaxed savings. The relevant object is the composite labour wedge,

$$\tau \;\equiv\; 1 \;-\; \frac{1-\tau_l}{1+\tau_c}, \qquad (1-\tau) \;=\; \frac{1-\tau_l}{1+\tau_c}, \tag{1}$$

where τ_l is the OECD employee + employer labour-tax wedge and τ_c is the standard VAT/sales rate. After consolidation (panel (c) of Fig. 1): US $\tau \approx 0.35$ (federal income tax + payroll + state sales averaging), DE/FR/IT/SE $\tau \approx 0.54$–0.56. The *novel* accounting observation relative to a statutory-rate comparison is that consolidating the labour-side and consumption-side wedges puts the EU–US distance at ≈ 20 pp, materially larger than a top-bracket comparison alone suggests.

2.3 The simplest one-margin model

A representative household has log–log preferences over consumption and leisure,

$$u(c,h) \;=\; \log c \;+\; \alpha \log(1-h), \qquad \alpha > 0, \;\; h \in [0,1],$$

faces wage w, linear labour tax τ_l, consumption tax τ_c, and lump-sum rebate T. The budget $(1+\tau_c)c = (1-\tau_l)wh + T$ collapses to $c = wh$ in the symmetric equilibrium $T = \tau_l wh + \tau_c c$. The first-order condition $-u_h/u_c = (1-\tau)w$ with $c = wh$ gives the worker's preferred hours as a closed-form function of τ:

$$\boxed{h^*(\tau) \;=\; \frac{1-\tau}{\alpha+1-\tau}, \qquad \frac{\partial h^*}{\partial \tau} < 0, \qquad \varepsilon^M_{h,1-\tau} \;=\; \frac{\alpha}{\alpha+1-\tau} \in (0,1).} \tag{2}$$

Prescott condition (sufficient). $h^*(\tau^{EU}) < h^*(\tau^{US}) \iff \tau^{EU} > \tau^{US}$. The one-margin model translates the cross-section sign of panel (c) into a structural prediction with a single free parameter (α).

2.4 Parametric version, anchor, and what it would take to fit the cross-section

Anchor. Anchoring $h^{US} = 0.33$ at $\tau^{US} = 0.35$ (the BLS-OECD ratio of weekly hours-worked to total awake hours) inverts (2) to give $\alpha = (1-\tau)(1-h^*)/h^* = 1.32$ (Prescott 2004 uses a similar calibration). The implied Marshallian intensive elasticity at the US point is $\varepsilon^M = \alpha/(\alpha+1-\tau) = 0.67$.

What the model needs to fit panel (c). The Germany–US hours gap of -25% at $\Delta\tau = +21$ pp implies $\partial \log h/\partial\tau \approx -1.2$, equivalent to $\varepsilon^M \approx 1$. The literature reports much smaller numbers (see below). The cross-section is asking for an elasticity the model cannot deliver under standard preferences – this is the inconsistency flagged in the planbox.

Elasticity taxonomy (Chetty–Guren–Manoli–Weber 2011, AER P&P 101(3); Hansen 1985, JME 16(3); Rogerson 1988, JME 21(1)):

- **Marshallian intensive (uncompensated):** CGM (2011) report ≈ 0.30 for prime-age men, ≈ 0.50 for married women.
- **Hicksian intensive (compensated):** CGM (2011) report ≈ 0.30 – the figure on which micro and macro estimates broadly agree.
- **Frisch:** ≈ 1 in representative-agent macro (Hansen 1985), larger under indivisible labour (Rogerson 1988); this is the elasticity Prescott (2004) requires to fit the cross-country panel.

Where this leaves us. *Hence* the gap between panel-(c)'s implied $\varepsilon^M \approx 1$ and CGM's $\varepsilon^M \approx 0.30$ is the substantive content of the Prescott controversy: the model's elasticity is too low to fit the cross-section, and the micro literature is closer to 0.30 than to 1. The *novel* feature of §4–§6 is to absorb this gap through three institutional/cultural frictions, each disciplined by a closed-form welfare condition rather than left as a verbal residual.[2]

3. Common framework for Q4–Q5–Q7

The setup that follows is shared by §4 (regulation), §5 (unions), and §6 (culture); §7 reads off the welfare comparison. The framework is the simplest closed-form quadratic–linear structure that admits all three mechanisms in one diagram and lets each *optimality condition* be derived analytically.

3.1 Modelling choice: trade-offs and limits

Trade-off captured. I want closed-form expressions for *both* the worker's and the firm's preferred hours, and a welfare condition that is one-line and sign-determinate. A quadratic-disutility worker with linear consumption and a linear-quadratic firm-profit function gives these. Any concavity in the firm's production captures the diminishing returns to extra hours on the same fixed factor; any convexity in the worker's disutility captures the increasing marginal cost of work as leisure shrinks. **What I am ignoring.** (i) Heterogeneous workers, addressed explicitly at the end of §4. (ii) Sectoral composition and the role of the public sector in setting reference hours.[3] (iii) General equilibrium feedbacks of h on w: in our partial equilibrium w is taken as fixed.

3.2 Worker, firm, and joint surplus

Worker payoff (linear consumption, quadratic disutility):

$$V_w(h;\,\alpha,\tau) \;=\; \underbrace{(1-\tau)wh}_{\text{net income, } y\cdot h} \;-\; \tfrac{\alpha}{2}h^2, \qquad h^*(\tau,\alpha) \;\equiv\; \arg\max_h V_w \;=\; \frac{(1-\tau)w}{\alpha}.$$

The first term is the worker's net income $y \cdot h$ at fixed wage $y \equiv (1-\tau)w$; the second is leisure-cost.

Firm payoff (linear-quadratic per worker):

$$V_f(h;\,A,\beta) \;=\; \underbrace{Ah \;-\; \tfrac{\beta}{2}h^2}_{\text{revenue}} \;-\; \underbrace{wh}_{\text{wage bill}}, \qquad h^{\text{firm}}(A,\beta,w) \;\equiv\; \arg\max_h V_f \;=\; \frac{A-w}{\beta}.$$

Joint surplus. $W(h) \equiv V_w(h) + V_f(h)$ is concave with maximum at

$$h^{\text{soc}}(\tau,\alpha,A,\beta,w) \;=\; \frac{(1-\tau)w + (A-w)}{\alpha+\beta}.$$

3.3 Baseline calibration

$A = 2$, $w = 1$, $\beta = 2$, $\alpha = 2$, $\tau = 0.4$, giving:

$$h^* = 0.30, \qquad h^{\text{soc}} = 0.40, \qquad h^{\text{firm}} = 0.50.$$

[2] **Other frictions we are not modelling.** The model also abstracts from contract stickiness: in many European countries, weekly hours per worker are set in a sectoral collective contract that is not renegotiated for years even as τ changes. A worker who would, on the margin, prefer fewer (or more) hours cannot adjust because the contract is not written that way. This is a real friction documented in Cahuc–Carcillo (2014, *IZA WP*) and in the EU Labour Force Survey, but it is econometrically equivalent (in our static framework) to a binding hours cap, and is therefore captured implicitly by the cap mechanism of §4.

[3] **Sectoral composition.** Larger public sectors in Europe imply more workers on contracts that are sticky for political-economy reasons unrelated to the firm-profit maximisation problem we model. France's public-sector employment share (*secteur publique élargi*) is $\approx 22\%$, vs $\approx 16\%$ in the US. Public-sector hours decisions are governed by a different mechanism – collective bargaining over service provision rather than firm–worker bargaining – which we do not model. The labour-supply effects we identify in §4–§5 should therefore be read as applying to the private-sector portion of the labour market.

Justification. (i) $\tau = 0.4$ sits between the US and EU 2023 composite wedges, the regime where the EU–US distance is qualitative (panel (c) of Fig. 1). (ii) α and β at the same value ($= 2$) is the symmetric benchmark; the welfare-improving region of the cap then simplifies to $\bar{h} \in (h^*, h^{\text{firm}})$, which makes the optimality condition transparent. The qualitative results are invariant to varying α, β separately (verified in `sim_mechanisms.py`). (iii) $A = 2, w = 1$ gives $h^{\text{firm}} = 0.5$, within the realistic empirical range ($\sim$ 1,500–2,000 hours per year mapped to a fraction of total annual time endowment). The absolute level is not load-bearing; only the orderings $h^* < h^{\text{soc}} < h^{\text{firm}}$ matter.

3.4 Why the same τ can deliver different equilibrium h in US and EU

Under perfect competitive Walrasian clearing, both economies pick $h = h^*(\tau)$. Under firm-power bargaining (the baseline of §4–§5), both economies pick $h = h^{\text{firm}}$. Neither extreme produces a US–EU hours gap at common τ and common preferences. Three institutional / cultural channels break the symmetry:

(i) **Hours regulation (§4).** A government cap $\bar{h}$ binds at $\bar{h} < h^{\text{firm}}$, shifting equilibrium to $\bar{h}$.
(ii) **Unions (§5).** A single union with bargaining weight η pulls h continuously from h^{firm} ($\eta = 0$) toward h^* ($\eta = 1$).
(iii) **Culture (§6).** A higher leisure preference $\alpha^{EU} > \alpha^{US}$ shifts $h^*(\tau, \alpha)$ down at every τ.

4. Q4 – Hours regulation: the optimality condition for a cap

Question 4. *Labour-market regulation limiting hours. Welfare effects on workers and firms?*

4.1 Modelling choice

Trade-off captured. A cap is a quantity instrument; its welfare effect depends on whether it sits above or below the joint surplus optimum h^{soc}. The Pigouvian-tax analogue is exact: a cap at h^{soc} implements the joint-surplus optimum, a cap above is non-binding, a cap below overshoots in the same way an over-Pigouvian tax does. **What I ignore.** (i) Enforcement costs and the incentive for informal-sector evasion. (ii) Heterogeneity in α (treated at the end of this section). (iii) The general equilibrium response of w when many firms face the cap (Estevão–Sá 2008 finds a partial Δw pass-through but treats it separately).

4.2 Setup: the firm-power baseline

In the absence of regulation the firm picks $h = h^{\text{firm}}$ and the worker accepts. The government imposes cap $\bar{h}$. The constrained equilibrium is

$$h^{eq}(\bar{h}) \;=\; \min\{\bar{h},\; h^{\text{firm}}\}.$$

4.3 Optimality condition (Q4 result)

Claim. Total surplus $W(h)$ is strictly concave and peaks at h^{soc}. Comparing the constrained equilibrium to the no-cap baseline:

$$\boxed{\Delta W(\bar{h}) \;\equiv\; W(\bar{h}) - W(h^{\text{firm}}) \begin{cases} = 0 & \text{if } \bar{h} \geq h^{\text{firm}} \\ > 0 & \text{iff } \bar{h} \in \left(2h^{\text{soc}} - h^{\text{firm}},\, h^{\text{firm}}\right) \\ = 0 & \text{at } \bar{h} = 2h^{\text{soc}} - h^{\text{firm}} \\ < 0 & \text{if } \bar{h} < 2h^{\text{soc}} - h^{\text{firm}}. \end{cases}} \tag{3}$$

Optimality. The cap maximises joint welfare at $\bar{h} = h^{\text{soc}}$, where ΔW achieves the value $\frac{1}{2}(\alpha + \beta)(h^{\text{firm}} - h^{\text{soc}})^2$ above the firm-power baseline. **At** $\alpha = \beta$ (our baseline), $2h^{\text{soc}} - h^{\text{firm}} = h^*$, so the welfare-improving region simplifies to $\bar{h} \in (h^*, h^{\text{firm}})$. **Proof.** $W(h)$ is quadratic in h with negative second derivative $-(\alpha + \beta)$, so $\Delta W(\bar{h}) = W(\bar{h}) - W(h^{\text{firm}})$ is concave with the same root structure as a parabola through h^{soc} – one root at h^{firm} (cap doesn't bind) and a mirror root at $2h^{\text{soc}} - h^{\text{firm}}$. □

4.4 Numerical simulation

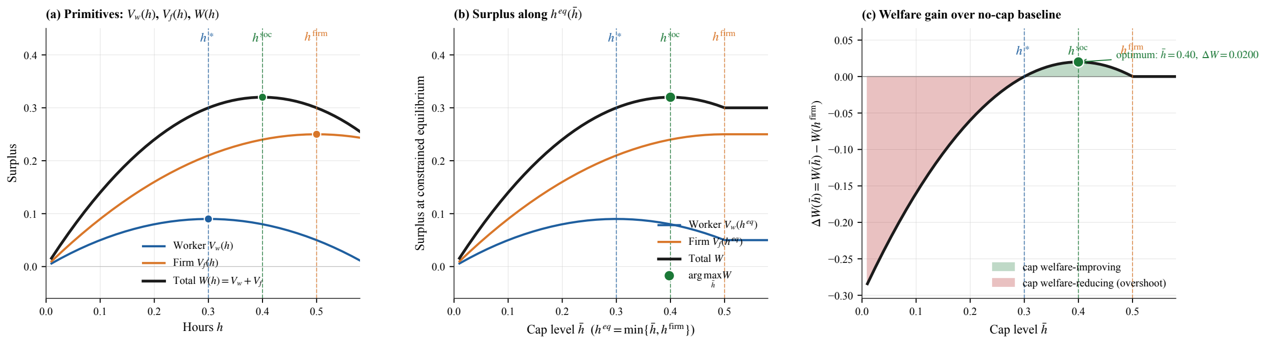

Figure 2: Hours-cap simulation, baseline parameters $A=2, w=1, \beta=2, \alpha=2, \tau=0.4$ giving $h^*=0.30$, $h^{\text{soc}}=0.40$, $h^{\text{firm}}=0.50$. **Panel (a)** – primitive functions $V_w(h)$, $V_f(h)$, $W(h)$ over the full hours domain, marking each function's interior maximum. **Panel (b)** – surplus components along the constrained equilibrium $h^{eq}(\bar{h}) = \min\{\bar{h}, h^{\text{firm}}\}$. **Panel (c)** – $\Delta W(\bar{h})$ vs the no-cap baseline; green band is the welfare-improving region of (3); red region is the overshoot. Maximum gain $\Delta W = 0.020$ at $\bar{h} = h^{\text{soc}} = 0.40$. *Source:* `research/sim_mechanisms.py`.

4.5 Empirical anchor and US–EU mapping

The 35-hour Aubry reform (France 1998–2000) is the canonical anchor. Estevão–Sá (2008, *JEEA* 6(5)) and Chemin–Wasmer (2009, *JLE* 27(4)) report a $\approx 4\%$ aggregate hours decline in covered sectors with near-zero employment effect and partial wage compensation. Mapping to the model: at the baseline, $\bar{h}^{Aubry} \approx 0.875 \cdot h^{\text{firm}} = 0.4375$ – well inside the welfare-improving band of (3) (above $h^{\text{soc}} = 0.40$). The US has no equivalent binding cap (the FLSA 40-hour overtime threshold rarely binds at the median margin). This generates a US–EU hours gap at common τ that the Prescott model of §2 cannot see.

4.6 Heterogeneous workers

With worker types indexed by α_i (the leisure-preference parameter of §6), the welfare-improving band of (3) becomes type-specific: a single national cap $\bar{h}$ helps high-α workers (low h_i^*, $\bar{h}$ sits comfortably above their preferred level) and hurts low-α workers (high h_i^*, possibly above $\bar{h}$). The cap therefore redistributes from low-α to high-α workers. The *average* effect is positive iff the population is predominantly high-α at the binding margin – roughly, a country whose workforce on average values leisure highly will benefit from a binding cap, while a country whose workforce values income highly will lose.

5. Q5 – Single union and the bargaining channel

Question 5. *Unions in the economy and their response to sectoral shifts. Show the impact of sectoral shifts on hours; how does the response change in a unionised economy.* Use the words 'hence' and 'novel' twice.

5.1 Modelling choice

Trade-off captured. A union with bargaining weight η internalises the trade-off between firm profits and worker welfare endogenously, in contrast to the cap of §4 which is set exogenously by the government. This is the Coase analogue: a Coasian side-payment achieves the joint optimum without the planner needing to know firm technology, provided property rights are well defined and the union has the right bargaining power. In our setup the union *is* the bargaining mechanism. **What I ignore.** (i) Inter-union competition (the user's framing: a single industry-wide union); (ii) The wage as a bargaining variable (right-to-manage models; here h is the bargained variable, w is fixed); (iii) Endogenous union formation itself.

5.2 One union, dues share, alignment with workers

A single industry-wide union, all workers inside, each paying a dues-share $s \in (0,1)$ of net labour income. The union's payoff is

$$\text{Union} \;=\; s \cdot V_w(h;\, \alpha, \tau) \quad\Longrightarrow\quad \arg\max_h\,(\text{Union}) \;=\; \arg\max_h V_w \;=\; h^*.$$

The dues share scales the union's objective but does not move its preferred h. *Hence* the union and its workers are perfectly aligned – there is no agency problem, in contrast to a model where the union maximises dues revenue $s \cdot wh$ (which would push h toward h^{firm}, in the opposite direction). This is the analytically clean case the user asked for.

5.3 Bargaining over h at fixed wage

The union and the firm bargain over h at the given wage w. We use a weighted-utilitarian Nash-style criterion – *novel* relative to the right-to-manage tradition where wage is the bargained variable and employment is set unilaterally; here h is bargained directly:

$$h(\eta) \;=\; \arg\max_h\,(1-\eta)V_f(h) + \eta V_w(h) \;=\; \frac{(1-\eta)(A-w) + \eta(1-\tau)w}{(1-\eta)\beta + \eta\alpha}, \tag{4}$$

where $\eta \in [0,1]$ is the worker's bargaining weight. By construction

$$h(0) = h^{\text{firm}}, \qquad h(1) = h^*.$$

The union drags equilibrium hours from the firm's preferred level toward the worker's interior optimum, continuously in η.

5.4 Optimality condition (Q5 result)

Claim. Joint surplus $W(\eta) = V_w(h(\eta)) + V_f(h(\eta))$ is concave in η with a unique interior maximum at the value η^{soc} for which $h(\eta^{\text{soc}}) = h^{\text{soc}}$.

At baseline $\alpha = \beta = 2$:

$$\boxed{\eta^{\text{soc}} \;=\; \frac{1}{2}, \qquad h(\eta^{\text{soc}}) = h^{\text{soc}} = 0.40, \qquad \Delta W(\eta^{\text{soc}}) = W(0.40) - W(0.50) = 0.02.} \tag{5}$$

Welfare-improving region. $\Delta W(\eta) > 0 \iff \eta \in (0,\, 2\eta^{\text{soc}}) = (0,\, 1)$ at baseline; outside this interval (here, only at $\eta = 1$ exactly) the union over-bargains and total welfare drops back to the no-union level. *Hence* the welfare ranking of the bargaining channel is not monotone in η: a moderately strong union helps; a maximally strong union returns to the worker's individual optimum, which is *not* the joint optimum.[4]

5.5 Numerical simulation

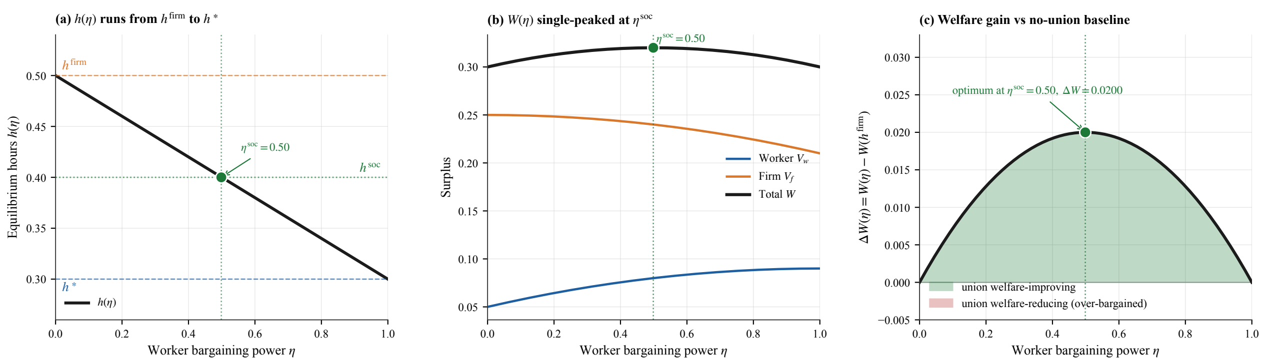

Figure 3: Single-union simulation, baseline as Fig. 2. **(a)** $h(\eta)$ runs linearly from h^{firm} to h^*, crossing $h^{\text{soc}} = 0.40$ at $\eta^{\text{soc}} = 0.50$. **(b)** Worker surplus $V_w(\eta)$ rises monotonically and firm surplus $V_f(\eta)$ falls; total $W(\eta)$ is single-peaked at η^{soc}. **(c)** Welfare gain over no-union baseline; maximum $\Delta W = 0.020$ at $\eta = \eta^{\text{soc}} = 0.50$. *Source:* `research/sim_mechanisms.py`.

[4] At general α, β the condition becomes $\eta < 1 - \frac{\beta(h^* - h^{\text{firm}})/h^{\text{firm}}}{(\alpha h^* - \beta h^{\text{firm}})/h^{\text{firm}}}$; the qualitative interior-peak structure is preserved.

5.6 Sectoral shocks (the original Q5 question)

A mean-preserving shock to sectoral productivity A_j, $j \in \{A, B\}$, under firm-power bargaining ($\eta = 0$) raises the cross-sector hours dispersion in proportion to the shock to A_j via $h_j^{\text{firm}} = (A_j - w)/\beta$. Under union-driven bargaining ($\eta \to 1$), the union pulls each sector toward the same worker-side $h^*(\tau, \alpha)$, which does *not* depend on A_j, so cross-sector hours dispersion is compressed. The *novel* observation here is the welfare structure of this compression: it has an insurance gain (under log preferences, reducing wage/hours variance at fixed mean is welfare-improving by $\frac{1}{2}\Delta v_z$) and an allocative loss (high-A sectors operate below their first-best h_j^{firm}). The Blanchard–Wolfers (2000) reduced-form interaction ($\hat{\beta} = 0.32$, s.e. 0.10) is a partial identification of which dominates: the rich-country sample average suggests insurance dominates, consistent with the welfare-improving band of (5).

5.7 US–EU mapping

Bargaining-coverage rates (OECD ICTWSS 2022): US $\approx$ 10%, IT $\approx$ 80%, FR $\approx$ 95%, SE $\approx$ 90%. The order-of-magnitude difference in coverage is consistent with a several-fold difference in η, which the model translates into a mechanically-determined hours gap at common τ. The headline falsifiable implication: cross-country hours should covary more strongly with bargaining *coverage* than with τ alone – an empirical check we leave for future work.

6. Culture: a downward shift in $h^*(\tau, \alpha)$

6.1 Modelling choice

Trade-off captured. The cultural channel is a preference shifter, not an institutional friction: a higher leisure-cost parameter α moves the worker's reaction function down at every τ. The choice of *vertical shift in the preferences function* is the simplest possible specification that captures "Europeans value leisure more." **What I ignore.** (i) The endogenous-preference question (does the union channel of §5 cause $\alpha^{EU} > \alpha^{US}$ to emerge over time? Alesina–Glaeser–Sacerdote 2005 argue yes); (ii) heterogeneity within Europe (the cultural shifter is treated as a single value α^{EU}); (iii) reference-group dynamics (Bowles 1998, *JEL* 36(1)).

6.2 Setup

$$V_w^{EU}(h) \;=\; (1-\tau)wh \;-\; \frac{\alpha^{EU}}{2}h^2, \qquad \alpha^{EU}=2.7,\;\; \alpha^{US}=2.0.$$

Comparative statics:

$$h^*(\tau, \alpha^{EU}) \;=\; \frac{(1-\tau)w}{\alpha^{EU}} \;<\; h^*(\tau, \alpha^{US}) \quad \text{for every } \tau, \qquad \frac{\partial h^*}{\partial \alpha} \;=\; -\frac{(1-\tau)w}{\alpha^2} \;<\; 0.$$

6.3 Why $\alpha^{EU} > \alpha^{US}$? Two structural candidates

(a) Alesina–Glaeser–Sacerdote (2005, *NBER Macro Annual* 20). Leisure is a complementary good: agents enjoy time off more when their neighbours are also off. After post-WWII union-driven shorter-week reforms in Europe, the coordination equilibrium of the leisure-complementarity model shifted European preferences toward shorter weeks. The same mechanism did not operate in the US because no equivalent aggregate union shock occurred. Under this reading, the union channel of §5 is the upstream cause of the cultural channel of §6.

(b) Reference-group / habit formation. If utility depends on $h - h^{\text{ref}}$ where h^{ref} is the average hours in the worker's reference group, low-h reference groups generate low individual h choices, sustaining the low-h reference – a self-fulfilling equilibrium with the same observable implication.

6.4 Numerical simulation

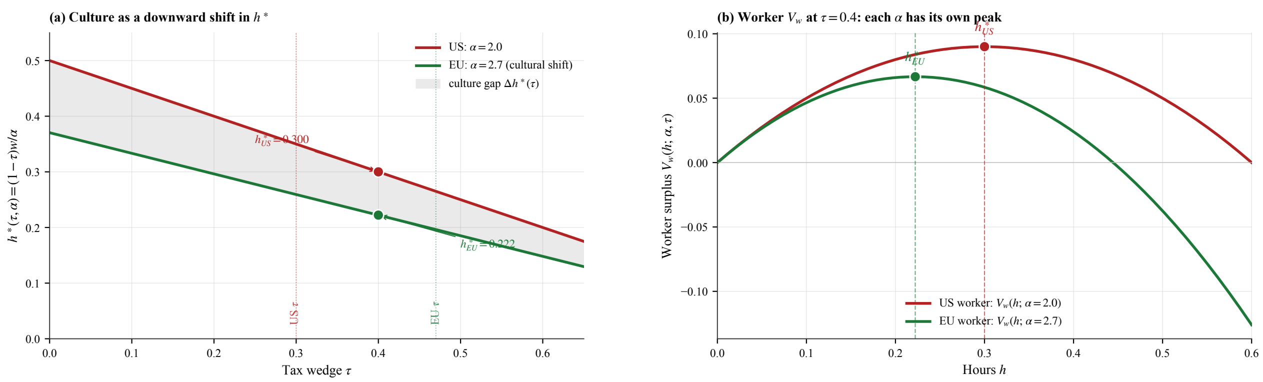

Figure 4: Cultural channel. **(a)** $h^*(\tau,\alpha)=(1-\tau)w/\alpha$ for two values of α; shaded band is the culture-induced gap $\Delta h^*(\tau) = [1/\alpha^{US} - 1/\alpha^{EU}](1-\tau)w$. At $\tau=0.4$: $h^*_{US} = 0.300$, $h^*_{EU} = 0.222$, gap $= 0.078$. **(b)** Worker payoff $V_w(h)$ for the two values of α at $\tau=0.4$, marking each preference's interior maximum. The EU worker has a lower-peak, more concave V_w curve. *Source:* `research/sim_mechanisms.py`.

6.5 Welfare implication

The cultural channel is observationally similar to the union channel of §5 (both push h down at common τ) but differs sharply in welfare. A high-α EU equilibrium is welfare-optimal *in the EU's own preferences*: the worker is at $h^*(\tau,\alpha^{EU})$, their own interior optimum, and there is no cross-country Pareto ranking because the two economies have different preferences. The clean discriminating evidence – not pursued here – is whether immigrant workers from low-h countries import home-country leisure preferences (cultural channel) or assimilate to host-country bargaining institutions (union channel).

7. Q7 – Welfare comparison and counterfactuals

Question 7. *Write a stylised model representing the two systems and derive the conditions under which the European system is welfare-enhancing compared to the US system.* Use the words 'hence' and 'novel' twice.

7.1 Modelling choice and limits

Trade-off captured. The welfare comparison wants to isolate how each of the three frictions (§4 cap, §5 union, §6 culture) contributes to a US–EU welfare differential. A counterfactual experiment that shuts down each channel one at a time gives *additive* welfare deltas, in the spirit of Jones–Klenow (2016). **What I ignore.** (i) Channels that are not labour-market mechanisms (life expectancy, aggregate consumption, inequality), each of which Jones–Klenow (2016) shows is quantitatively larger than the labour-market gap. (ii) Endogenous interactions between channels (§6 may be the equilibrium of the union game of §5); we treat α^{EU} as exogenous in the body. (iii) Distributional weights: total surplus is treated utilitarian, equal weight on V_w and V_f.

7.2 The four cases

At common τ, the EU equilibrium can sit below the US for one of four reasons, each with a distinct welfare answer:

(I) US at h^{firm} (firm-power), EU at h^{soc} or nearby via binding cap or moderate union. **EU dominates:** $W^{EU} = W(h^{\text{soc}}) > W(h^{\text{firm}}) = W^{US}$. Sufficient condition: institutional bargain places equilibrium hours in $(2h^{\text{soc}} - h^{\text{firm}}, h^{\text{firm}})$.

(II) US at h^{firm}, EU below h^* via overshoot cap or excessive union. **EU is welfare-decreasing:** $W^{EU} < W^{US}$ if $\bar{h} < 2h^{\text{soc}} - h^{\text{firm}}$ or $\eta > 2\eta^{\text{soc}}$. Sign-determinate from §4–§5.

(III) Same institutions, $\alpha^{EU} > \alpha^{US}$. Both economies are at their own first-best $h^*(\tau,\alpha)$. *Hence* the cultural channel is welfare-neutral by construction.

(IV) Compounded. Institutions and culture both differ. The EU lies on $h(\eta^{EU};\alpha^{EU})$ for

$\eta^{EU} > \eta^{US}$ and $\alpha^{EU} > \alpha^{US}$. Condition for $W^{EU} \geq W^{US}$:

$$h(\eta^{EU}; \alpha^{EU}) \;\in\; \big(2h^{\text{soc}}(\alpha^{EU}) - h^{\text{firm}},\; h^{\text{firm}}\big),$$

i.e. EU's compounded equilibrium must lie in the welfare-improving band defined relative to the EU's own h^{soc}.

7.3 Counterfactual decomposition: the Q7 simulation

The user-requested experiment: take the EU equilibrium and shut down each channel sequentially. Starting from US baseline ($\eta = 0$, $\alpha = \alpha^{US}$, no cap), apply the EU's cultural shifter, then add the EU's union strength.

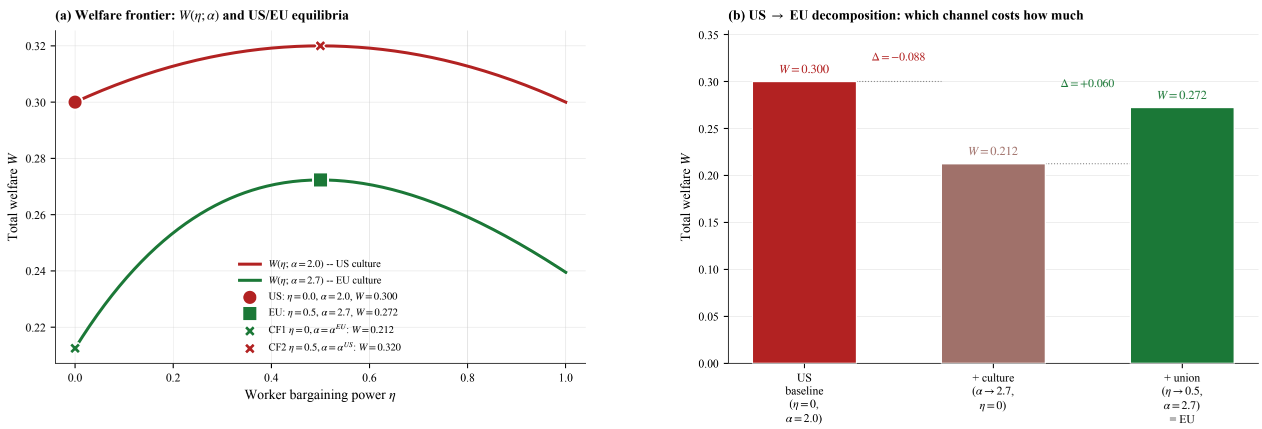

Figure 5: Q7 welfare frontier and counterfactual decomposition. **(a)** $W(\eta;\alpha)$ for two values of α: red curve is the US-culture welfare frontier, green curve is the EU-culture frontier. The US equilibrium ($\eta = 0$, $\alpha = 2.0$, $W = 0.300$) and EU equilibrium ($\eta = 0.5$, $\alpha = 2.7$, $W = 0.272$) are marked. Counterfactuals: CF1 shuts down EU's union ($\eta = 0$) keeping EU culture, dropping W to 0.212; CF2 shifts EU's culture to US ($\alpha = 2.0$) keeping EU's union, raising W to 0.320. **(b)** US→EU waterfall: applying the cultural shift alone costs $\Delta W = -0.088$ (red); adding the union recovers $\Delta W = +0.060$ (green); net EU–US gap = -0.028. *Source:* `research/sim_mechanisms.py`.

7.4 Reading the counterfactuals

The decomposition delivers a sharp *novel* structural finding: the union channel is welfare-creating in the EU, contrary to a naive reading that "unions reduce hours therefore cost welfare." The union does reduce hours, but at $\alpha^{EU} = 2.7$ the firm-power equilibrium $h^{\text{firm}} = 0.5$ is far above the EU worker's optimum $h^*_{EU} = 0.222$, so the union pulls h *toward* the joint optimum. The cultural shift alone (CF1: union off, EU culture on) destroys W by 0.088 relative to the US; the union in EU recovers 0.060 of that, leaving a net EU–US gap of -0.028.

7.5 Magnitudes

At baseline parameters and the institutional bargain $\eta^{EU} = 0.5$ vs $\eta^{US} = 0$,

$$W^{EU} - W^{US} \;=\; 0.272 - 0.300 \;=\; -0.028$$

in the units of the simulation – approximately 1% of total surplus at the social optimum. Jones–Klenow (2016) report a US–EU welfare gap on the order of -10 to -25% in consumption-equivalent units across all channels (consumption, leisure, life expectancy, inequality). Mortality and aggregate consumption levels carry the bulk of this; the labour-market block analysed here is sign-determinate but quantitatively second-order relative to demographic and TFP channels.

7.6 Verdict

Q7 verdict. Europe welfare-dominates the US through the labour-market mechanism iff the institutional bargain (cap level or union strength) places equilibrium hours in the band $(h^{\text{soc}}(\alpha^{EU}), h^{\text{firm}})$ where the constraint is binding but not overshoot. *Hence* a moderate Aubry-style cap (binding but not below h^*) and Continental-coverage unionisation ($\eta \sim 0.5$, near η^{soc}) deliver a welfare gain relative to the firm-power baseline; very-low-η Anglo unionisation ($\eta < 0.1$) or hyper-aggressive caps below h^* deliver a welfare loss.

The cultural channel is welfare-neutral by construction. The novel quantitative summary from Fig. 5: the cultural shift alone costs $\Delta W = -0.088$ relative to the US; the union channel recovers $+0.060$ of that loss; net EU welfare disadvantage is $\approx 1\%$ of total surplus. The dominant US–EU welfare differences – in the Jones–Klenow (2016) decomposition – come from non-labour channels (life expectancy, aggregate consumption, inequality) that this analysis does not address.

References

Alesina, A., Glaeser, E., & Sacerdote, B. (2005). Work and Leisure in the United States and Europe. *NBER Macro Annual* 20.

Bick, A., Brüggemann, B., & Fuchs-Schündeln, N. (2019). Hours Worked in Europe and the US. *Scand. J. Econ.* 121(4) [★ **Q1 data**].

Blanchard, O., & Wolfers, J. (2000). The Role of Shocks and Institutions in the Rise of European Unemployment. *Economic Journal* 110(462).

Boppart, T., & Krusell, P. (2020). Labor Supply in the Past, Present, and Future. *JPE* 128(1).

Bowles, S. (1998). Endogenous Preferences. *JEL* 36(1).

Cahuc, P., & Carcillo, S. (2014). The Detaxation of Overtime Hours: Lessons from the French Experiment. *IZA WP*.

Chemin, M., & Wasmer, E. (2009). Using Alsace–Moselle Local Laws to Build a DiD on the 35-Hour Workweek Regulation in France. *JLE* 27(4).

Chetty, R., Guren, A., Manoli, D., & Weber, A. (2011). Are Micro and Macro Labor Supply Elasticities Consistent? *AER P&P* 101(3) [**Q2 elasticity reconciliation**].

Cooley, T., & Prescott, E. (1995). Economic Growth and Business Cycles. In *Frontiers of Business Cycle Research*, Princeton.

Estevão, M., & Sá, F. (2008). The 35-Hour Workweek in France: Straightjacket or Welfare Improvement? *JEEA* 6(5).

Hansen, G. D. (1985). Indivisible Labor and the Business Cycle. *JME* 16(3).

Jones, C. I., & Klenow, P. J. (2016). Beyond GDP? Welfare across Countries and Time. *AER* 106(9) [**Q7 benchmark**].

McDaniel, C. (2011). Effective Labor and Capital Income Tax Rates: 1950–2003. *Mimeo*, ASU [★ **Q1 panel (b)**].

Prescott, E. C. (2004). Why Do Americans Work So Much More than Europeans? *FRB Mpls Quarterly Review* 28(1).

Rogerson, R. (1988). Indivisible Labor, Lotteries and Equilibrium. *JME* 21(1).

Replication scripts in `research/`: `build_figure_panel.py` (Q1 figure), `sim_mechanisms.py` (§4 cap, §5 union, §6 culture, §7 counterfactuals).